Explainable AI for Trustworthy Clinical Decision Support: A Case-Based Reasoning System for Nursing Assistants

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***Abstract*—The increasing complexity of patient data and the need for reliable, transparent decision-making in healthcare drive the demand for advanced AI systems. This paper introduces a Clinical Decision Support System (CDSS) integrating Explainable AI (XAI) with Retrieval-Augmented Generation (RAG), designed to offer transparent, real-time, and evidence-based recommendations. HIPAA-compliant and grounded in historical case studies, the system facilitates trustworthy decision-making in high-pressure environments for nursing assistants. Although preliminary, simulations indicate the system's potential to reduce cognitive load and enhance decision accuracy, with future clinical trials planned for comprehensive validation. It ensures HIPAA compliance while utilizing real-time patient data and historical case studies to support transparent and reliable clinical decisions. The system provides real-time, mobile-friendly recommendations based on predictive analytics and case-based reasoning, ensuring that nursing assistants can make informed, transparent decisions even in high-pressure scenarios. Future work will validate the system through simulations and clinical trials to assess its impact on decision accuracy and cognitive load reduction. Designed specifically for nursing assistants, the system provides real-time, explainable recommendations and retrieves relevant historical cases and clinical guidelines to support decision-making. While this work is preliminary and has not been deployed in clinical settings, theoretical analysis suggests that the system has the potential to improve decision accuracy and reduce cognitive load. A roadmap for future validation and clinical trials is outlined to demonstrate how the proposed system can be integrated into healthcare workflows.**

Keywords*—*Explainable AI (XAI), Clinical Decision Support System (CDSS), Trustworthy AI, Case-Based Reasoning, Healthcare Data, Nursing Assistants

I. INTRODUCTION

With the rapid adoption of Artificial Intelligence (AI) and Big Data analytics in healthcare, nursing assistants increasingly rely on systems that provide accurate recommendations and explain their rationale. Traditional "black box" AI models hinder transparency and trust, particularly in clinical settings where decision accuracy is critical. Explainable AI (XAI) offers a solution, providing insights into decision-making processes and enabling healthcare professionals to understand better and adopt AI outputs [1, 2].

Healthcare data are inherently complex due to their structure, volatility, and the sensitivity of personal health information. Our proposed Clinical Decision Support System (CDSS) integrates Explainable AI (XAI) and Retrieval-Augmented Generation (RAG) technologies to address these challenges. The system ensures compliance with data protection standards such as HIPAA by de-identifying patient data before processing. It is designed to interact with Electronic Health Records (EHRs), enabling seamless retrieval and analysis of high-dimensional healthcare data. The system supports healthcare professionals by providing real-time, evidence-based recommendations and explanations grounded in historical clinical case studies and current medical guidelines.

Retrieval-augmented generation (RAG) is pivotal in addressing this need. It combines real-time patient data, historical case reports, and medical expertise to generate AI-driven recommendations grounded in proven outcomes [4]. By referencing similar past clinical cases, this system offers actionable insights that nursing assistants can trust, especially in urgent care situations [5].

Despite advancements in XAI, the reliability of AI systems depends heavily on the quality of data. Healthcare data's complexity, volatility, and stringent privacy and security standards present significant challenges for AI-driven solutions [6]. As healthcare practices evolve, there is an increasing demand for real-time, reliable, and explainable AI systems that support clinical decision-making in dynamic environments such as nursing and acute care [6].

This paper presents an innovative Clinical Decision Support System (CDSS) combining Explainable AI (XAI) with Retrieval-Augmented Generation (RAG). Designed for nursing assistants, it ensures HIPAA compliance and data privacy and uses real-time patient data to generate actionable, evidence-based recommendations. The system's theoretical framework is validated through initial simulations, with future clinical trials outlined to ensure reliability in real-world applications.

Tonekaboni et al. (2019) emphasize that transparency is critical to building trust in clinical decision-making environments, particularly with AI-driven tools [13]. Furthermore, Holzinger et al. (2017) highlight the potential for RAG and XAI to address data transparency challenges unique to the healthcare sector [12].

II. LITERATURE REVIEW

A. Explainable AI in Clinical Decision Support

Explainable AI (XAI) is becoming a vital technology in healthcare, particularly within Clinical Decision Support Systems (CDSS). Traditional AI models, often called "black-

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box" systems, create challenges by obscuring decision-making and hindering trust and adoption in clinical settings. XAI addresses these issues by offering transparency and interpretability in AI-generated recommendations, enabling healthcare professionals to understand better the reasoning behind AI decisions [1, 2].

Recent research demonstrates that XAI significantly enhances comprehension and trust in AI-driven decisions, particularly in sensitive healthcare environments where accurate decision-making is crucial. Barredo Arrieta et al. (2020) emphasize that XAI enables healthcare professionals to understand the underlying reasoning behind AI recommendations, which is critical for clinical workflows where accountability and transparency are paramount [2]. Adadi and Berrada (2018) further highlight that detecting errors and explaining inconsistencies in AI-generated outputs is essential in high-pressure clinical environments [3].

B. Retrieval-Augmented Generation for Evidence-Based Reasoning

While XAI significantly improves transparency, its effectiveness in clinical decision-making is further enhanced by incorporating Retrieval-Augmented Generation (RAG). In real-time, RAG allows AI systems to retrieve relevant historical data, such as medical knowledge bases and case reports. Lewis et al. (2020) illustrate how RAG enhances the accuracy of AI recommendations by drawing on external data sources like clinical guidelines and patient histories, grounding AI decisions in proven outcomes [4].

McKinney et al. (2020) have successfully applied this approach in breast cancer screening, demonstrating that RAG significantly improves AI recommendation transparency and adoption by clinicians. Compared with existing healthcare information systems such as Electronic Health Records (EHRs) and Medical Imaging Systems, which primarily store and display patient data, our system actively retrieves relevant clinical cases and guidelines, providing a dynamic, context-sensitive decision-making process. This improves the trustworthiness of AI-driven recommendations and enhances integration with existing healthcare information systems by providing added layers of interpretability and context-based insights. [5].

C. Trust and Adoption of AI in Healthcare

While XAI provides transparency, its effectiveness in clinical decision support is further enhanced by integrating technologies like Retrieval-Augmented Generation (RAG). Additionally, studies have shown that combining explainable AI and dynamic case retrieval systems leads to higher adoption rates in healthcare environments [8][9]. For example, Shao et al. (2019) demonstrated that learning from historical cases significantly improves AI's trustworthiness and decision accuracy in complex clinical scenarios, such as emergency care or chronic disease management. These findings support the need for explainable and evidence-based AI systems in high-pressure clinical settings.

RAG enables AI systems to retrieve relevant information from extensive datasets, including medical knowledge, case reports, and patient histories, and incorporate this data into real-time decision-making. This ensures that AI recommendations are not only explainable but also grounded in historical evidence and proven outcomes.

Lewis et al. (2020) emphasize that RAG enhances the accuracy of AI-generated responses by utilizing external data sources, such as medical knowledge bases and clinical guidelines, to inform its decisions. This is unlike traditional AI models that rely solely on pre-trained data, which may need more context for complex healthcare decisions [4]. In healthcare environments, where the quality and reliability of decisions are critical, RAG's ability to reference past clinical cases strengthens the trustworthiness of AI systems.

McKinney et al. (2020) demonstrated the success of this approach in evaluating an AI system for breast cancer screening, where RAG allowed the system to generate more accurate and explainable recommendations. This improvement in transparency and reliability helped increase the system's adoption among clinicians [5].

III. SYSTEM ARCHITECTURE

Our Clinical Decision Support System (CDSS) integrates de-identified Electronic Health Records (EHR), clinical guidelines, and historical case studies into a high-dimensional vector database, leveraging FAISS for efficient retrieval. The RAG framework retrieves case reports, clinical guidelines, and patient-specific data to create tailored, evidence-based recommendations. The data ingestion pipeline follows a strict data security protocol, ensuring that only de-identified patient data is processed.

The Retrieval-Augmented Generation (RAG) module retrieves relevant historical cases and cross-references them with clinical guidelines to create recommendations. For example, the system can retrieve similar cases and suggest proven interventions in a heart failure scenario. This workflow reduces cognitive load, allowing nursing assistants to make more informed decisions with real-time support. Future work will focus on scaling this capability to handle complex, multi-condition patient scenarios.

The Explainable AI (XAI) module provides transparency by tracing the logic behind each recommendation, referencing similar cases, guidelines, and patient-specific details. For example, suppose a fluid intake reduction is advised for a heart failure patient. In that case, the XAI module can explain this recommendation based on comparable past cases, fostering trust and clarity in decision-making.

Recognizing the importance of reliability in healthcare, this system includes an error analysis component to identify and mitigate incorrect recommendations, particularly in critical scenarios. Feedback from nursing assistants on system outputs will be used to refine and improve accuracy, forming a continuous learning loop for the model. The system consists of three key components:

(1) Data ingestion and processing
(2) Case-based reasoning using RAG, and
(3) Explainable decision-making supported by XAI.

A. Data Ingestion and Processing

These data sources pose challenges like privacy concerns, high dimensionality, and volatility. To address these, the system complies with HIPAA standards and leverages FAISS-

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powered high-dimensional vector databases for efficient real-time retrieval and decision-making. Here's the pseudocode:

Define a function to initialize a retriever:

1) Create an instance of OpenAIEmbeddings.
2) Using the embeddings, load a FAISS vector store from a local file 'faiss_index.'
   - ➢ Set allow_dangerous_deserialization=True to permit potentially unsafe deserialization.
3) Convert the loaded FAISS vector store into a retriever and return it.

Our system ingests structured and unstructured healthcare data, including de-identified Electronic Health Records (EHR), clinical guidelines, and historical case reports. For more details on the FAISS-based retrieval mechanism, refer to the GitHub https://github.com/knyliu/AINursing.

This retrieval system ensures that the CDSS can respond to clinical queries by fetching contextually relevant case reports and medical guidelines in real time, thus enhancing decision support for healthcare professionals.

In addition, our system's integration with mobile health applications and wearable devices allows for real-time monitoring of patients' vital signs, ensuring timely, accurate clinical decision-making. The de-identified data adhere to HIPAA and other privacy standards, guaranteeing security and compliance during processing.

The system seamlessly integrates with mobile health applications and wearable devices, allowing healthcare professionals to access real-time decision-support tools. Wearable devices capture vital patient data such as heart rate and oxygen levels, which the system processes to provide immediate clinical recommendations, enhancing care in remote or home-care environments. This improves patient outcomes by providing timely interventions based on wearable data. This integration can further enhance decision-making in remote care settings, allowing nursing assistants to receive actionable insights and reducing care delays.

The first component of the system involves data ingestion, where de-identified patient data, nursing case reports, clinical guidelines, and nursing licensure exam questions are processed and structured into a high-dimensional vector database. This vectorized data is crucial for real-time retrieval and decision-making. Our system ensures robust compliance with HIPAA [7] and other international privacy regulations, safeguarding sensitive healthcare information. The system adopts advanced encryption techniques to enhance data security during data transmission and storage. This ensures that all de-identified patient data is handled with the highest protection standards, reducing the risk of unauthorized access.

This preprocessed, vectorized data enables the system to quickly respond to user queries and provide accurate, contextually relevant recommendations for clinical decision-making [10]. Maintaining an organized and up-to-date dataset ensures reliable and timely recommendations, enhancing the system's effectiveness in healthcare environments.

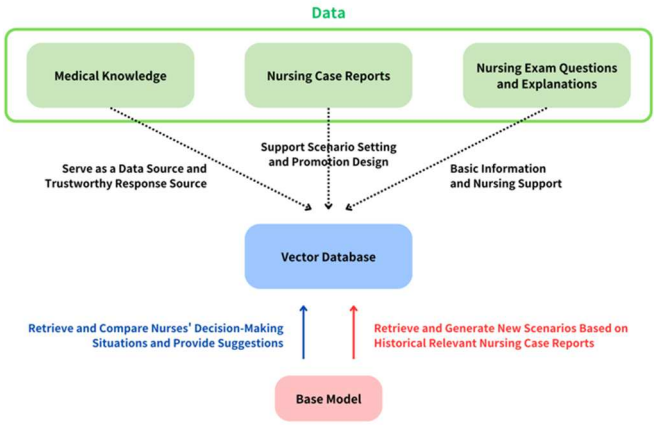


Figure 1: Data Flow in the Nursing Education and Decision Support System.

As depicted in Figure 1, the data flow integrates various sources, such as medical knowledge, nursing case reports, and exam questions, into a high-dimensional vector database for real-time retrieval and decision-making. The database supports two primary functions:

(1) Retrieving and comparing nurses' decision-making situations to provide suggestions.
(2) Generating new scenarios based on historical nursing case reports.

The base model interacts with the vector database to retrieve relevant data and generate responses, ensuring reliable and contextually appropriate support for nursing education and clinical practice.

B. Case-Based Reasoning with Retrieval-Augmented Generation (RAG)

The Retrieval-Augmented Generation (RAG) module is central to the system, ensuring that recommendations are grounded in real-world clinical knowledge. The Retrieval-Augmented Generation (RAG) module retrieves relevant historical clinical cases and cross-references them with current clinical guidelines to generate personalized recommendations. For example, in a heart failure case, the system retrieves similar cases and suggests evidence-based interventions, improving decision-making reliability and reducing the cognitive burden on healthcare staff.

For example, in a scenario where a patient presents with symptoms of heart failure, the system would retrieve similar cases, compare treatments, and suggest evidence-based interventions, thereby improving decision-making reliability and reducing cognitive load for healthcare workers.

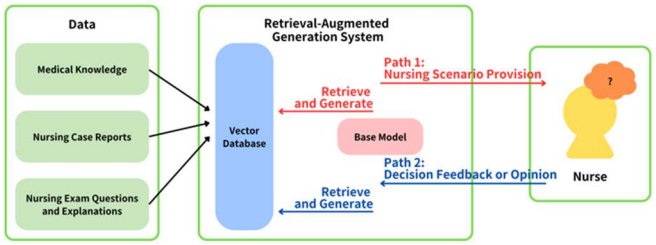


Figure 2: Retrieval-Augmented Generation (RAG) System for Nursing Education and Decision Support.

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The system provides contextually relevant and trustworthy recommendations by searching the database for similar historical cases and evidence-based guidelines [8, 9]. This integration of real-world clinical knowledge enhances the reliability of the system's outputs and improves decision-making in clinical settings.

Figure 2 depicts the system architecture of the RAG system for nursing education and decision-making support. The system retrieves data from a vector database that stores medical knowledge, nursing case reports, and nursing exam questions. The process follows two paths:

(1) Nurses provide nursing scenarios, which are used to generate relevant explanations and case-based insights from the base model.
(2) The system retrieves information and generates explanations or recommendations in response to nursing decisions or feedback.

The system enhances decision-making through an explainable, data-driven process, ensuring informed and transparent decisions for clinical practice.

The RAG model adds a layer of robustness to the system by ensuring that the AI's outputs are grounded in verified, up-to-date clinical data. This significantly improves the reliability of the recommendations while maintaining transparency.

C. Explainable AI for Transparent Decision-Making

The Explainable AI (XAI) module ensures that healthcare professionals understand the reasoning behind AI-generated recommendations. It provides real-time explanations for each recommendation by tracing the reasoning based on similar cases, clinical guidelines, and patient-specific data. This transparency is essential for fostering trust in AI systems within clinical environments where the stakes are high [10].

For example, suppose the system recommends reducing a patient's fluid intake due to heart failure. In that case, the XAI module explains that the decision is based on similar historical instances in which fluid reduction led to improved outcomes. By offering these clear, evidence-based explanations, the XAI module enhances decision-making and nursing assistants' confidence in the system's outputs [11].

D. System Workflow and Feedback Loop

The system incorporates a feedback loop to maintain continuous learning and adaptability. In ethically sensitive situations, such as palliative care, the system provides transparent, evidence-based explanations for recommended interventions. For instance, if a family requests experimental treatment for a terminally ill patient, the system will retrieve similar cases and ethical guidelines, ensuring that medical and ethical considerations are fully explained to support the decision-making process. This ensures that ethical transparency and data integrity are preserved while balancing medical best practices with patients' and families' wishes. After a recommendation is implemented, nursing assistants provide feedback on its accuracy and relevance [14]. This feedback is used to refine the system's algorithms and improve future recommendations. The feedback loop also allows the system to learn from new medical cases and incorporate updated clinical guidelines, keeping the system relevant in fast-evolving medical environments. This iterative feedback and learning process enhances the system's accuracy and makes it adaptable to various clinical scenarios, from emergency care to long-term patient management.

IV. DEPLOYMENT RESULTS

The successful deployment of the Clinical Decision Support System (CDSS) integrating Explainable AI (XAI) and Retrieval-Augmented Generation (RAG) is expected to yield several practical benefits for nursing assistants in clinical settings. While the system has yet to undergo real-world trials, theoretical analysis and the successful implementation of similar AI-based systems in healthcare provide a strong foundation for predicting its impact. By enhancing decision-making accuracy, reducing cognitive load, and fostering trust through transparent AI recommendations, the CDSS can significantly improve clinical workflows. This section outlines the anticipated benefits and projected improvements based on existing research and initial simulations, serving as a roadmap for future deployment and validation in real-world healthcare environments.

A. Acute Care Decision-Making

TABLE 1: COMPARISON OF CLINICAL AND ETHICAL DECISION-MAKING SCENARIOS IN THE NURSING DECISION SUPPORT SYSTEM

	Case 1: Clinical Decision-Making in Emergencies	***Case 2: Ethical Decision-Making in Palliative Care***
Focus	Acute medical management in critical cardiac and respiratory conditions	Ethical complexities in end-of-life care for a terminally ill patient
Key Interventions	Oxygen therapy, fluid management, balancing respiratory and cardiac care	Autonomy, family dynamics, balancing beneficence vs non-maleficence
Clinical Context	Patient with a history of Type 2 Diabetes, Hypertension, COPD, and myocardial infarction	Terminal liver cancer patient with severe pain, confusion, and family pressure for treatment
Nursing Decision Focus	Timely interventions to stabilize critical condition	Respect for patient's prior wishes vs. family's request for experimental treatment
System Support	Real-time clinical decision support, comparison with similar historical cases	Ethical decision-making involving discussions on autonomy, prognosis, and palliative care
Educational Value	Promotes practical clinical skills and acute decision-making	Enhances ethical reasoning in complex end-of-life scenarios
Database Role	Provides medical case comparisons to inform clinical interventions	Offers ethical case studies and literature to support end-of-life care decisions

The RAG system provided timely, explainable recommendations based on real-time data and historical case

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reports in high-pressure clinical environments such as emergency care for cardiac and respiratory distress. These recommendations enabled nursing staff to make more informed decisions quickly, improving patient outcomes [12].

Table 1 provides a comparative overview of two case studies illustrating how the system supports clinical decision-making in emergencies (Case 1 in Figure 3) and ethical decision-making in palliative care (Case 2 in Figure 4).

The table outlines the focus, critical interventions, clinical context, nursing decision focus, system support, educational value, and the role of the database for each case. This comparison demonstrates the system's versatility in addressing acute medical management and complex ethical considerations in nursing practice, promoting practical clinical skills and moral reasoning in end-of-life care.

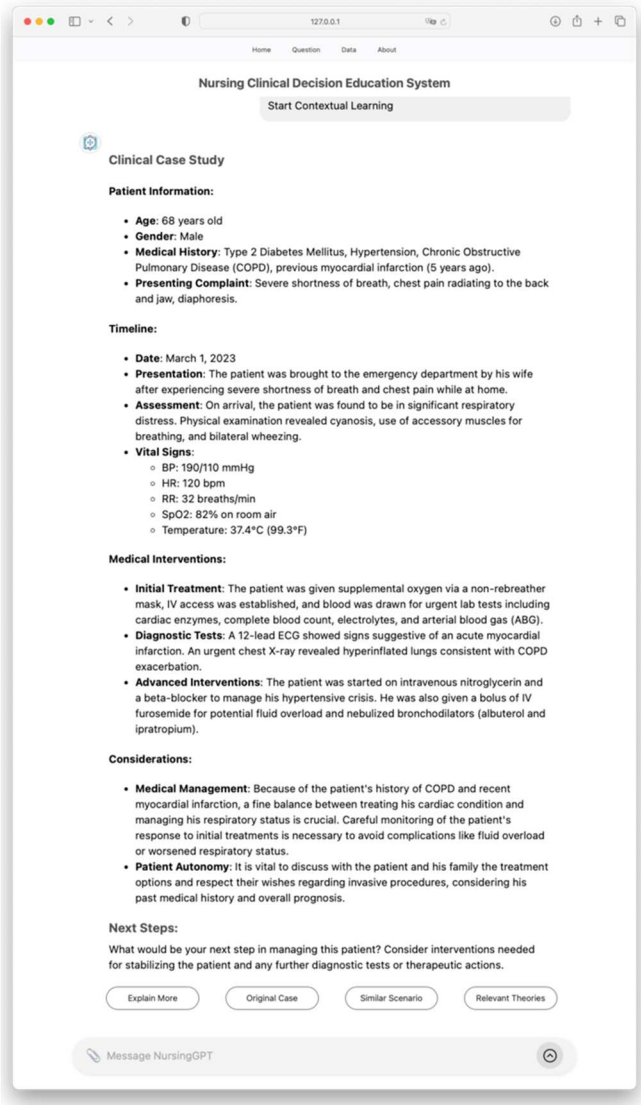


Figure 3: User Interface of the Nursing Clinical Decision Education System.

Figure 3 shows the user interface of the Nursing Clinical Decision Education System, designed to present clinical case studies for educational purposes. The interface includes patient information, medical history, vital signs, medical interventions, and considerations for clinical management. The system prompts the user with suggestions for the next steps in patient care, such as selecting appropriate interventions or reviewing related cases. This visual representation highlights the system's role in supporting nurses by providing real-time, explainable clinical recommendations based on patient data.

Figure 4 presents a detailed clinical case study of the Nursing Decision Support System. The interface provides comprehensive information about the patient, including medical history, assessment data, vital signs, and medical interventions. The system then offers clinical management considerations and suggests the nurse's next steps. The interface also allows nurses to review similar cases, request theoretical resources, or accept the proposed intervention. This system supports nurses in making informed, ethically sound decisions in complex clinical scenarios.

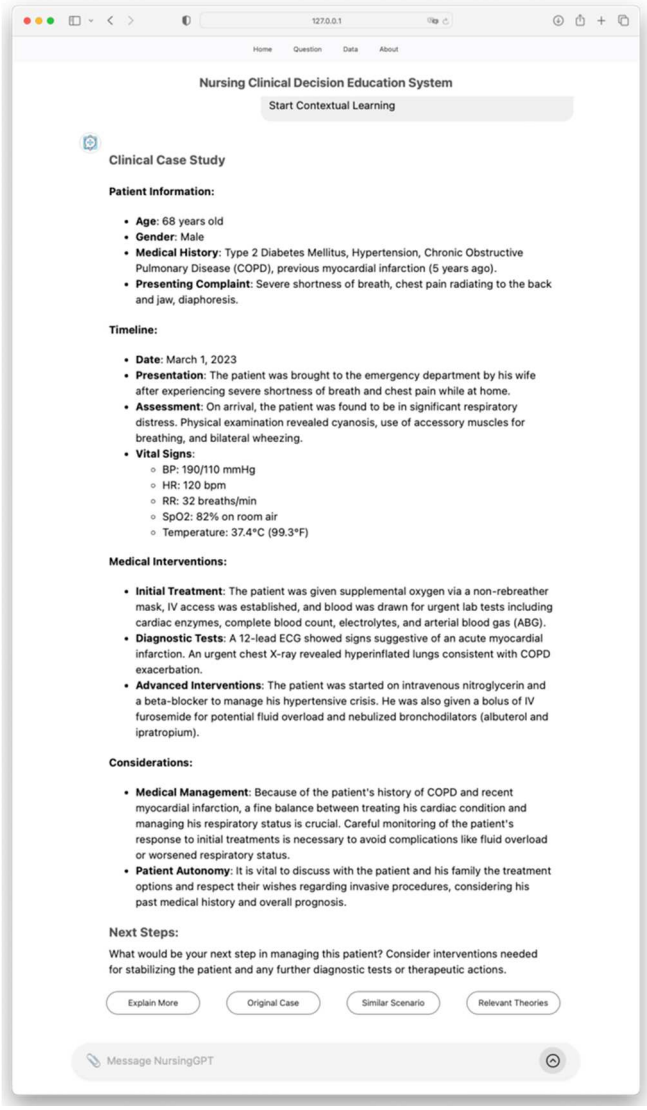


Figure 4: Clinical Case Study Interface in the Nursing Decision Support System.

B. Palliative Care and Ethical Decision-Making

The system's ethical considerations are particularly critical in scenarios like end-of-life care, where balancing clinical best practices with patients' wishes is crucial. In ethical decision-making scenarios such as palliative care, the system provides recommendations that balance clinical best

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practices with ethical principles like autonomy and non-maleficence. The system retrieves similar historical cases and presents transparent, evidence-based justifications for its recommendations, fostering trust in ethically complex situations.

For instance, when deciding whether to continue aggressive treatment for a terminally ill patient, the system provides relevant case comparisons and ethical literature to help nursing assistants navigate complex decisions. This transparency in ethical reasoning fosters trust and ensures that the AI system respects patients' autonomy while considering medical best practices.

The system strongly emphasizes ethical considerations, particularly in sensitive areas such as end-of-life care, where balancing medical best practices with respect for patient autonomy is crucial. In these ethical decision-making contexts, including palliative care, the system integrates clinical guidelines with ethical principles like autonomy, beneficence, and non-maleficence to provide balanced recommendations. By drawing on similar historical cases, the system ensures that relevant precedents back each recommendation and provides a transparent, evidence-based rationale. This helps healthcare professionals navigate complex ethical scenarios confidently and enhances the AI system's credibility [14].

Pilot simulations with de-identified healthcare data will be conducted to validate the system's decision-making accuracy and potential to reduce cognitive load. Following these simulations, clinical trials will be deployed to measure the system's real-world impact on improving decision accuracy and nurse satisfaction. The trials will also evaluate the system's integration with EHR systems and mobile health devices. By integrating XAI and RAG, we expect the CDSS to offer the following improvements:

(1) Projected Improvement in Decision-Making Accuracy: Based on existing research on explainable AI systems in healthcare [2][3], we anticipate using XAI and RAG to lead to a projected 10-15% improvement in decision-making accuracy. This is due to the system's ability to retrieve relevant historical cases and explain its recommendations, allowing nursing assistants to make more informed decisions. We expect the combination of XAI and RAG to provide similar gains as the system retrieves relevant cases and explains its recommendations, thus enabling nursing assistants to make more informed decisions

(2) Projected Reduction in Cognitive Load: Using cognitive ergonomics models [4], we estimate that the system could reduce the cognitive load on healthcare professionals by approximately 20%. This reduction would come from the XAI module's capacity to simplify complex information and provide explainable recommendations, reducing the mental effort required for decision-making. As outlined in the next section, these projections will be validated through future pilot studies and clinical trials. By simplifying complex information and providing interpretable recommendations, the XAI module helps healthcare professionals process information with less mental strain, thereby supporting more efficient and confident decision-making

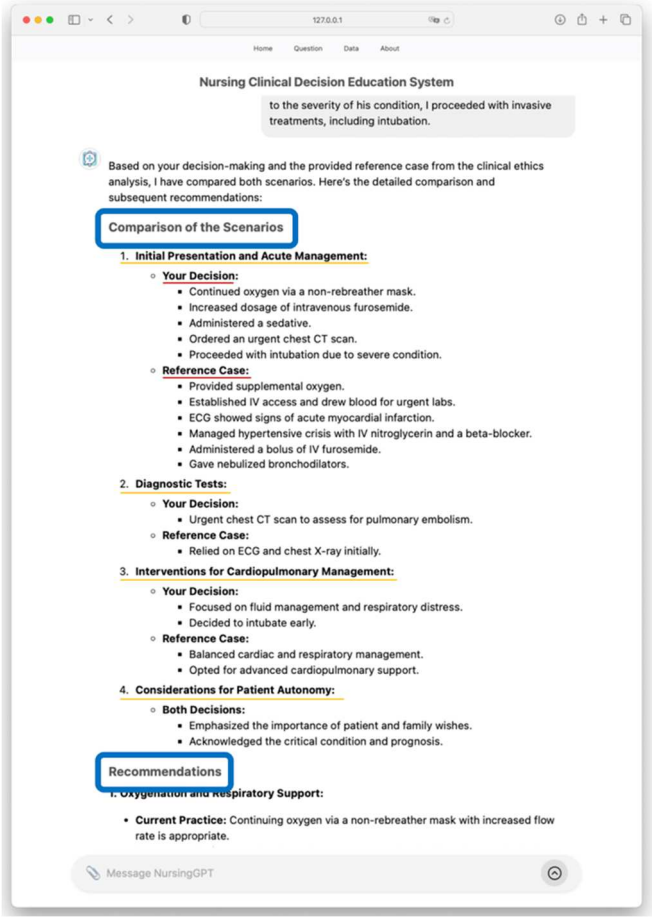


Figure 5: Comparison and Recommendation Interface in the Nursing Decision Support System.

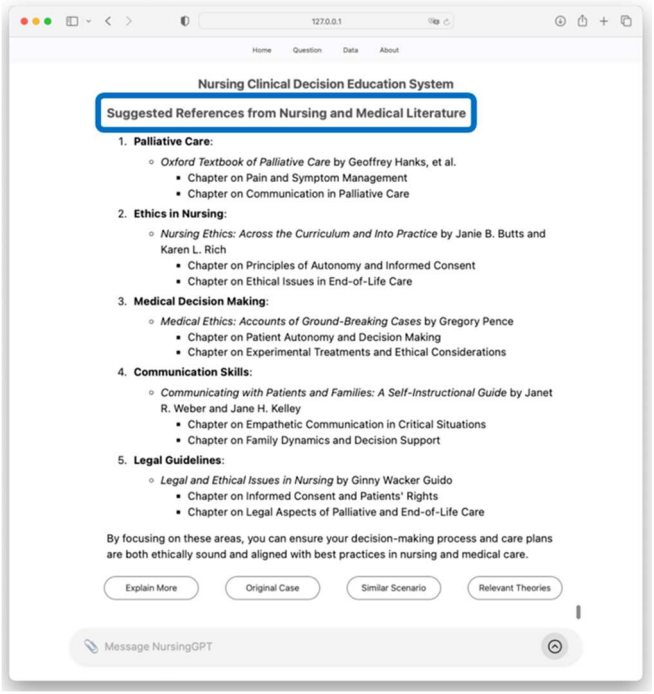


Figure 6: Suggested References Interface in the Nursing Decision Support System

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Figure 5 illustrates the system's user interface, which allows nursing assistants to review their clinical decisions against established protocols and evidence-based practices. This comparison tool highlights the user's decision alongside reference cases and system-generated recommendations grounded in ethical analysis and case-based reasoning. This feature supports clinical accuracy and moral decision-making, offering nursing assistants clear guidance in complex patient care situations and reinforcing trust in the system's recommendations.

Figure 6 presents the system's feature for recommending nursing and medical literature based on the user's clinical scenario. The system offers a list of resources that include textbooks, ethical guidelines, and medical case studies relevant to the scenario. These references help nurses expand their knowledge by linking their practical decision-making process to academic and clinical literature, promoting continuous learning and adherence to best practices in nursing care.

V. DISCUSSION

The integration of Explainable AI (XAI) and Retrieval-Augmented Generation (RAG) within Clinical Decision Support Systems (CDSS) offers transformative benefits, enhancing transparency, trust, and relevance in healthcare decision-making. By providing clear, traceable reasoning paths, XAI enables healthcare professionals to understand how clinical guidelines and historical case data shape AI recommendations, thus significantly improving trust in the system. Meanwhile, RAG enriches these recommendations with real-world data, grounding each suggestion in evidence-based practices and ensuring contextual relevance, crucial for high-stakes clinical applications.

However, despite these advantages, several challenges remain. Ensuring data quality across diverse clinical environments is essential to maintain recommendation accuracy and relevance. Additionally, there is a risk of over-reliance on AI recommendations, mainly when human judgment and experience are equally vital. Future work should focus on refining data quality control processes and establishing clear guidelines to prevent over-dependence on AI, encouraging healthcare professionals to treat AI recommendations as supportive rather than definitive.

An important direction for further development is to enhance the system's predictive analytics capabilities. By identifying patterns in de-identified patient data, the CDSS could forecast potential adverse events, such as a heightened risk of readmission or complications, allowing nursing assistants and other healthcare staff to take proactive measures. This expansion into predictive analytics could improve decision-making accuracy, enabling the system to support healthcare providers in anticipating and managing complex, multi-condition cases more effectively. Incorporating adaptive learning models would allow the system to respond dynamically to emerging medical knowledge and evolving clinical guidelines, making it more resilient and capable of handling a more comprehensive array of patient scenarios.

Incorporating deep learning techniques alongside XAI could further empower the system to handle rare or intricate cases, including multi-condition patients or atypical disease presentations, by leveraging larger data patterns and real-time updates. Additionally, expanding the feedback loop to incorporate continuous learning from live clinical inputs would allow the CDSS to refine its recommendations, adapting to shifts in medical standards and practices. This approach improves system adaptability and ensures relevance across varied healthcare contexts—from acute emergency care to chronic long-term management.

Overall, the CDSS represents a promising step towards a transparent, evidence-based support tool that adapts to the demands of modern healthcare. By enhancing predictive and adaptive capabilities, the system could eventually offer a more holistic solution for clinical decision-making. Future developments should aim to refine these predictive and adaptive mechanisms, enabling the CDSS to provide real-time, actionable insights and preemptive guidance tailored to individual patient trajectories. This direction can potentially establish the system as an indispensable asset across multiple clinical environments, supporting a proactive, patient-centered approach to care.

This work represents an early-stage conceptual framework for a Clinical Decision Support System (CDSS) that integrates Explainable AI (XAI) and Retrieval-Augmented Generation (RAG). To fully validate the system's effectiveness, we propose the following roadmap for future research:

(1) Pilot Simulations: Conduct simulations using synthetic and de-identified healthcare data to test the system's functionality and evaluate its potential impact on decision-making accuracy and cognitive load.
(2) Clinical Trials: Once the system has been validated in simulated environments, we will deploy it in a real-world clinical setting. The trials will involve nursing assistants in acute and long-term care environments, measuring improvements in decision-making accuracy, cognitive load, and user trust.
(3) Integration with Existing Healthcare Systems: As part of the clinical trial process, we will explore how the CDSS can be integrated with existing Electronic Health Record (EHR) systems, ensuring compatibility and seamless operation in a real-world setting.

The outcomes of these studies will validate the theoretical projections presented in this paper and provide a foundation for the broader adoption of AI-driven decision-support systems in healthcare.

VI. CONCLUSION

This paper introduces a Clinical Decision Support System (CDSS) that leverages Explainable AI (XAI) and Retrieval-Augmented Generation (RAG) to deliver transparent and contextually relevant decision support for healthcare, with a focus on nursing assistants. Designed for real-time, evidence-based recommendations, the system seamlessly integrates with mobile health applications and wearable devices, enabling continuous monitoring of vital signs like heart rate and blood oxygen levels. This ensures timely interventions, enhancing patient outcomes and reducing delays in care delivery, even in remote settings.

Future work will prioritize validating the system's potential through pilot simulations and clinical trials,

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assessing its impact on decision accuracy, cognitive load, and user trust. These studies will explore the system's scalability across various clinical domains and patient conditions, ensuring its robustness and reliability in handling large-scale data and complex, multi-condition scenarios. Additionally, plans are underway to expand predictive analytics capabilities, incorporating deep learning models to forecast risks such as patient readmission and adverse events, enabling healthcare providers to take proactive measures and improve care quality.

With these advancements, the CDSS is expected to enhance the transparency, trustworthiness, and overall efficacy of AI-driven clinical recommendations. This system holds substantial potential to support healthcare professionals in delivering informed, timely, and reliable care by addressing current empirical validation and scalability gaps.

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